

Project Design Phase
Problem ? Solution Fit Template

15 February 2025	10 May 2026
Team ID	HDI Predictor Team
Project name	HDI Predictor using Machine Learning and Flask
2 Marks	2 Marks

Problem ? Solution Fit Template:

The HDI Predictor solves the need to estimate Human Development Index values from measurable country indicators through a simple browser-based application.

Purpose:

The solution fits the problem because it turns country-level indicators into a reusable prediction workflow.

It reduces the need for manual comparison by combining preprocessing, regression modeling, and Flask presentation.

It improves usability by adding charts, confidence, and category output.

It supports a clear and consistent development workflow from dataset to demo.

- ☐ The solution fits the problem because it turns country-level indicators into a reusable prediction workflow.
- ☐ It reduces the need for manual comparison by combining preprocessing, regression modeling, and Flask presentation.
- ☐ It improves usability by adding charts, confidence, and category output.
- ☐ It supports a clear and consistent development workflow from dataset to demo.
- ☐ Template:

References:

Define CS, fit into CC	1. CUSTOMER SEGMENT(S) CS Who is your customer? I.e. working parents of 0-5 y.o. kids	6. CUSTOMER CONSTRAINTS CC What constraints prevent your customers from taking action or limit their choices of solutions? I.e. spending power, budget, no cash, network connection, available devices.	5. AVAILABLE SOLUTIONS AS Which solutions are available to the customers when they face the problem or need to get the job done? What have they tried in the past? What pros & cons do these solutions have? I.e. pen and paper is an alternative to digital notetaking	Explore AS, differentiate
	2. JOBS-TO-BE-DONE / PROBLEMS J&P Which jobs-to-be-done (or problems) do you address for your customers? There could be more than one, explore different sides.	9. PROBLEM ROOT CAUSE RC What is the real reason that this problem exists? What is the back story behind the need to do this job? I.e. customers have to do it because of the change in regulations.	7. BEHAVIOUR BE What does your customer do to address the problem and get the job done? I.e. directly related: find the right solar panel installer, calculate usage and benefits; Indirectly associated: customers spend free time on volunteering work (I.e. Greenpeace)	
Identify strong TR & EM	3. TRIGGERS TR What triggers customers to act? I.e. seeing their neighbour installing solar panels, reading about a more efficient solution in the news.	10. YOUR SOLUTION SL If you are working on an existing business, write down your current solution first, fill in the canvas, and check how much it fits reality. If you are working on a new business proposition, then keep it blank until you fill in the canvas and come up with a solution that fits within customer limitations, solves a problem and matches customer behaviour.	8. CHANNELS of BEHAVIOUR CH 8.1 ONLINE What kind of actions do customers take online? Extract online channels from #7	Extract online & offline CH of BE
	4. EMOTIONS: BEFORE / AFTER EM How do customers feel when they face a problem or a job and afterwards? I.e. lost, insecure > confident, in control - use it in your communication strategy & design.		8.2 OFFLINE What kind of actions do customers take offline? Extract offline channels from #7 and use them for customer development.	

The project context comes from the SmartBridge HDI Predictor handouts and project documentation.

1. <https://www.ideahackers.network/problem-solution-fit-canvas/>
2. <https://medium.com/@epicantus/problem-solution-fit-canvas-aa3dd59cb4fe>